

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1**(Adv-2021-P1-Model) **Date: 10-05-2026**

Time: 09.00Am to 12.00Pm **GTA-31** **Max. Marks: 180**

10-05-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2021-P1)_GTA-31_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCE-2021-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 10)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 11 – 16)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 23)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 24 – 29)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 30 – 35)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

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MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 42)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 43 – 48)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 49 – 54)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

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PHYSICS **Max Marks: 60**

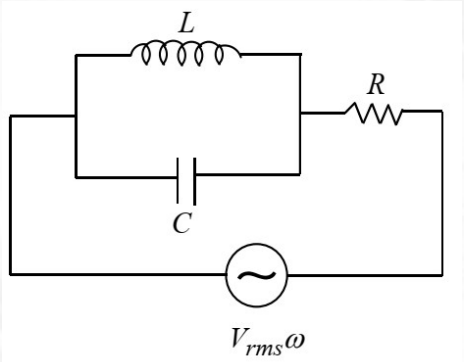
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases)

1. In the given L-C-R circuit the value of rms current through the AC source is

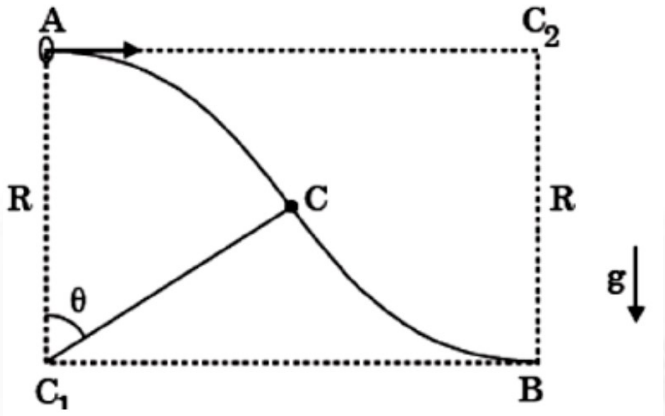


A) $\dfrac{V_{rms}}{\sqrt{R^2+\left(\omega L-\dfrac{1}{\omega C}\right)^2}}$ **B)** $\dfrac{V_{rms}}{\sqrt{R^2+\left(\dfrac{\omega L}{1-\omega^2 LC}\right)^2}}$

C) $\dfrac{V_{rms}}{\sqrt{R^2+\left(\omega^2 LC-\omega L\right)^2}}$ **D)** $\dfrac{V_{rms}}{\sqrt{R^2-\left(\omega^2 LC-\omega L\right)^2}}$

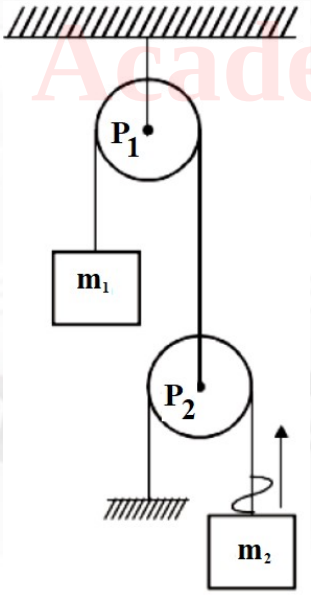
2. Smooth wire shaped as circular arc AC and CB are of same radius but centre of curvature are on opposite side as shown. A small bead is projected horizontally from A with speed such that normal force becomes zero just before point C. Speed of the particle when it reaches to B will be $\left[Take\,\theta=\cos^{-1}\left(\dfrac{5}{6}\right)\right]$

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A) $\sqrt{2gR}$ B) $\sqrt{3gR}$ C) $\sqrt{\frac{5}{2}gR}$ D) $\sqrt{\frac{3}{2}gR}$

3. In the figure shown all pulleys are light and there is no friction between pulley and strings. Masses $m_1 = m_2 = 2kg,$ and all strings have a linear mass density $\mu = 3\times10^{-2} kg/m,$ which makes masses of strings negligibly small compared to m_1 and m_2. Initially separation between pulley P_2 and mass m_2 is 20 cm . At this instant system is released from rest and a transverse pulse is created at the location of m_2



A) Pulse will reach P_2 in time 0.01s **B)** Pulse will reach P_2 in time 0.1s
C) Pulse with reach P_2 in time 1s **D)** Pulse will reach P_2 in time 10s

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4. A conducting wire of length L fixed at both ends is vibrating in its fundamental mode with angular frequency ω and maximum amplitude A. There exists a uniform and constant magnetic field of induction B perpendicular to the plane of oscillations of the wire. The maximum emf induced in the wire is

A) $\frac{BA\omega L}{\pi}$ **B)** $\frac{2BA\omega L}{\pi}$ **C)** $\frac{BA\omega L}{2\pi}$ **D)** $\frac{BA\omega L\pi}{2}$

SECTION 2

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +2** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Question Stem for Question Nos. 5 and 6

Question Stem

A rectangular box of length l consisting of two chambers of length $l/2$ each, separated by a smooth piston, lies on a horizontal surface with the piston being movable along its length. Each chamber consists of n moles of gas at pressure P_0, volume V_0 and temperature $T_0 = 127^0 C$. Initially the walls of the box are conducting and the piston is pushed slowly such that the temperature virtually remains constant. By the time the ratio of volumes of the chambers become 3 : 1 an amount of heat equal to Q is exchanged with the surroundings. On the other hand if walls are insulated and piston alone is conducting, the temperature of the gas becomes T, in the larger chamber by the time volume ratio becomes 3 : 1. Let C_p be equal to 2 R.

5. Then the magnitude of Q is $nRT_0 \ln x$ then x is

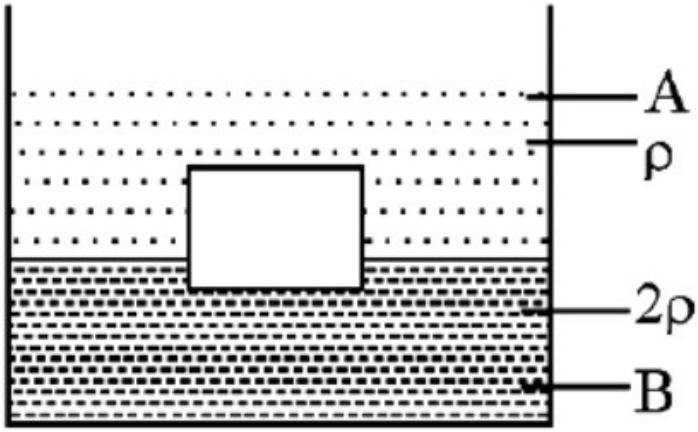
6. The value of T in $^\circ C$ is approximately

Question Stem for Question Nos. 7 and 8

Question Stem

A cubical block of side length 'a' is at equilibrium with one-third volume immersed in liquid B. Liquid A has density ρ and B has density 2ρ.

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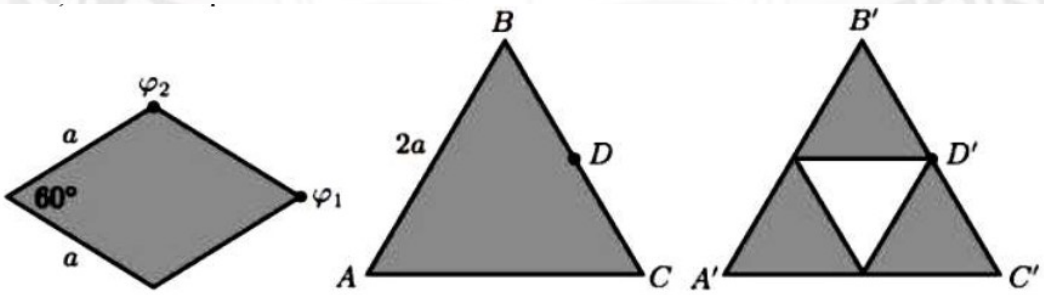
7. Density of block is $\frac{8\rho}{n}$, find n ?

8. If block is slightly pushed downward then time period of S.H.M is $2\pi\sqrt{\frac{8a}{ng}}$, find n ?

Question Stem for Question Nos. 9 and 10

Question Stem

A thin, flat, diamond-shaped dielectric plate with a side a and an acute angle $60°$ is charged uniformly with a surface charge density σ. The potential at the vertex of an acute angle of a rhombus is equal to φ_1, at the vertex of an obtuse angle is φ_2 (Fig. on the left). A thin plate in the shape of an equilateral triangle ABC with side $2a$ is cut out of the same dielectric and charged with the same surface charge density (Fig. in the centre). Note: all plates are removed from each other and other bodies.



9. The potential at the point D lying in the middle of the side of the triangular plate (middle figure) is $\varphi_D = \varphi_1 + p\varphi_2$. Then the value of p is ___

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10. Now an equilateral triangle with side a is removed from the triangular plate ABC (Fig. on the right). The remaining three shaded parts are all congruent equilateral triangles. The potential at the point C' of the plate with hole is $\varphi_{C'} = \left(\frac{q}{2}\varphi_2\right) - \varphi_1$. Then the value of q is

___

SECTION 3

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and

Choosing any other option(s) will get -2 marks.

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11. The universal gravitational constant is measured experimentally using Cavendish's experiment. This experiment is a very precise experiment which gives the value of G as $(6.6742 \pm 0.0010) \times 10^{-11} N.m^2 / kg^2$. Since the universal gravitational constant has some error in it, using it if we find escape speed and acceleration due to gravity on Earth's surface, our values will have error in it. (Assume mass of Earth $= 6.0000 \times 10^{24} kg$ & radius of Earth $= 6.4000 \times 10^6 m$). The measurement of mass of Earth has an error of 0.01% and radius of Earth has an error of 0.02%

Choose the CORRECT option(s)

A) Maximum percentage error in acceleration due to gravity $g = 0.065$

B) Maximum percentage error in acceleration due to gravity $g = 0.025$

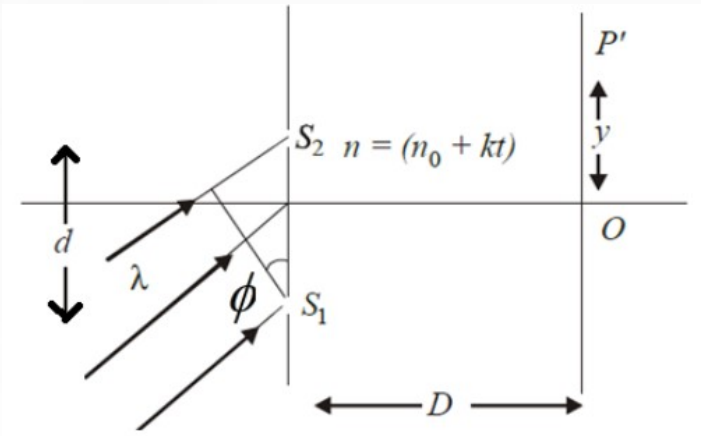
C) Value of acceleration due to gravity (approx. in m / s^2) $= 9.78$

D) Value of acceleration due to gravity (approx. in m / s^2) $= 8.78$

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12. For the Young's double slit experiment a monochromatic source light is used whose wavelength is λ strikes on the slits, separated by distance d as shown in the figure. In the figure n is refractive index of medium between slits and screen. Where n_0 and k are constants. $(D \gg d)$



A) The y co-ordinate of central maxima at any time t is $\frac{D\sin\phi}{n_0+kt}$

B) The velocity of central maxima at any time t as a function of time t is $\frac{-kD\sin\phi}{(n_0+kt)^2}$

C) If a glass plate of small thickness b is placed in front of S_1, its refractive index vary with time so that central maxima is formed at O is $n^1 = n_0 + kt + \frac{d\sin\phi}{b}$

D) Rate of change of velocity of the central maxima has no meaning as source is at large distance

13. Consider two light waves moving in z direction represented by two mutually perpendicular electric field vectors (with non-zero amplitudes): $E_x = A_x\cos(\omega_x t + \phi_x)$ and $E_y = A_y\sin(\omega_y t + \phi_y)$. Their superposition will result in a plane polarized light, if

A) $A_x = A_y, \phi_x = \frac{\pi}{2}, \phi_y = \pi$ **B)** $\omega_x = \omega_y, \phi_x = \frac{\pi}{2}, \phi_y = \pi$

C) $A_x = A_y, \omega_x = \omega_y, \phi_x = \frac{\pi}{2}, \phi_y = \pi$ **D)** $A_x \neq A_y, \omega_x = \omega_y, \phi_x = \frac{\pi}{2}, \phi_y = \pi$

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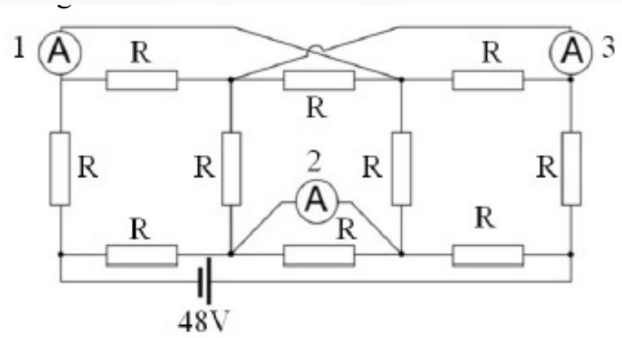
14. The schematic provided has all resistors with a resistance of three ohms, and the battery voltage is 48 volts. All ammeters are ideal.

Statement-1 : Readings of all ammeters is same.

Statement-2 : Reading of ammeter 1 is 5 A.

Statement-3 : Reading of ammeter 2 is 7A.

Statement-4 : Reading of ammeter 3 is 7A.



A) All statements are correct.

B) Only statement-1 is correct.

C) Statement-1 & statement-4 are correct.

D) Statement-2 & statement-3 are correct.

15. A particle of mass m is thrown in a parabolic path towards earth having mass M and radius R. It comes to greatest approach distance $R/2$ from surface of earth. Magnitude of total energy change required to make the particle moving in an elliptical path having eccentricity $\frac{1}{2}$, with the greatest approach point either apogee or perigee are:

A) $\frac{1}{3}\frac{GMm}{R}$ **B)** $\frac{1}{6}\frac{GMm}{R}$ **C)** $\frac{1}{2}\frac{GMm}{R}$ **D)** $\frac{GMm}{R}$

16. Electrons having de-Broglie wave length λ are incident on target in X-ray tube. Minimum wavelength of X-rays generated is λ_1. Then choose INCORRECT statement(s).

A) If λ is increased the value of λ_1 decreases.

B) If λ is increased the value of λ_1 increases.

C) The value of λ_1 will not change if target metal is changed.

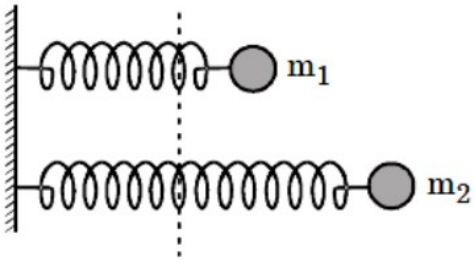
D) The value of λ_1 does not depend on λ.

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SECTION 4

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

17. The drawing shows a top view of a frictionless horizontal surface, where there are two springs with particles of mass m_1 and m_2 attached to them. Each spring has a spring constant of $1200 N/m$. The particles are pulled to the right and then released from the positions shown in the drawing. How much time passes before the particles are side by side for the first time at $x = 0m$ if $m_1 = 3kg$ and $m_2 = 27kg$. If time is $\frac{n\pi}{80}$sec, fill n in OMR sheet.



18. A particle of mass m and charge + q is projected from origin with velocity $\vec{v} = v_0\hat{i}$ in a magnetic field $\vec{B} = B_0 x\hat{k}$. Here V_0 and B_0 are positive constants of proper dimensions. The radius of curvature of the path of the particle when it reaches maximum positive x coordinate is $\sqrt{\alpha\frac{mv_0}{qB_0}}$. What is the value of 2α ?

19. A radioactive sample contains two radio nuclides A and B having decay constant λhr^{-1} and $2\lambda hr^{-1}$. initially 25% of total decay comes from A. How long (in hour) will it take before 75% of total decay comes from a. [take $\lambda = \ell n3$]

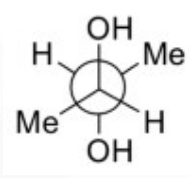
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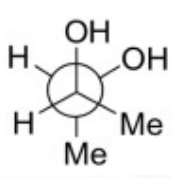
CHEMISTRY **Max. Marks: 60**

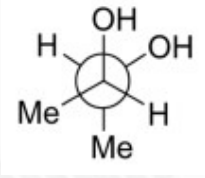
SECTION 1

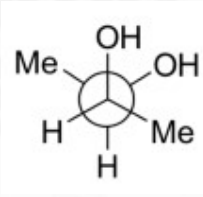
- This section contains **Four (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
- *Full Marks* : +3 If ONLY the correct option is chosen;
- *Zero Marks* : 0 If the none of the options is chosen (i.e. the question is unanswered);
- *Negative Marks* : -1 In all other cases.

20. Among the following, the conformation that corresponds to the most stable conformation of 2R, 3R – butane – 2, 3 – diol is

A) 

B) 

C) 

D) 

21. In the given question K_a values of five weak acids are given. K_a for $HA = 10^{-8}, HB = 10^{-6}, HC = 2\times10^{-8}, HD = 10^{-10}, HE = 10^{-7}$. Identify the incorrect statement.

A) Among NaA, NaB, NaC, NaD and NaE, NaD is most extensively hydrolysed.

B) A solution that is 1M with respect to each of the 5 acids will have a pH between 3 and 5.

C) pH of 0.1M aqueous solution of NaB is approximately 9.5.

D) Adding 0.1M of solution of HE to 0.01M solution of HB does not change its pH.

22. Match according to Bohr's model

Column – I		Column – II	
A)	Radius of nth orbit	P)	Inversely proportional of z
B)	Kinetic Energy of electron in n^{th} orbit	Q)	Integral multiple of $\frac{h}{2\pi}$
C)	Velocity of electron in n^{th} orbit	R)	Inversely Proportional to n^2
D)	Angular momentum of electron	S)	Inversely proportional to n
		T)	Proportional to Z^2

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A) A –P; B – R, T; C – S; D – Q

B) A – P; B – Q, T; C – S, R; D – P

C) A – S, T; B – S, T; C – P; D – R

D) A – R; B – R; C – T; D – S

23. If following ligand substitution reaction occurs:

$$\left[Cr(H_2O)_4(en)\right]^{3+} \xrightarrow[-2H_2O]{+2NH_3} \text{Complex cation X}$$

then complex cation X and $\left[Cr(H_2O)_4(en)\right]^{3+}$ do not differ in:

A) Colour **B)** Number of geometrical isomers.

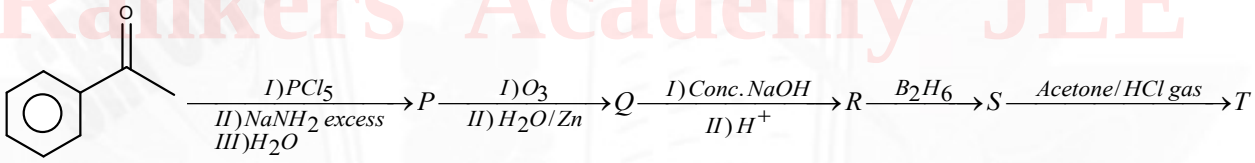
C) Number of optical isomers **D)** Spin only magnetic moment

SECTION 2

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +2** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks*** **: 0** In all other cases.

Question Stem for Question Nos. 24 and 25

Question Stem



24. Sum of the heteroatoms in the each molecule of P, Q, R, S and T is

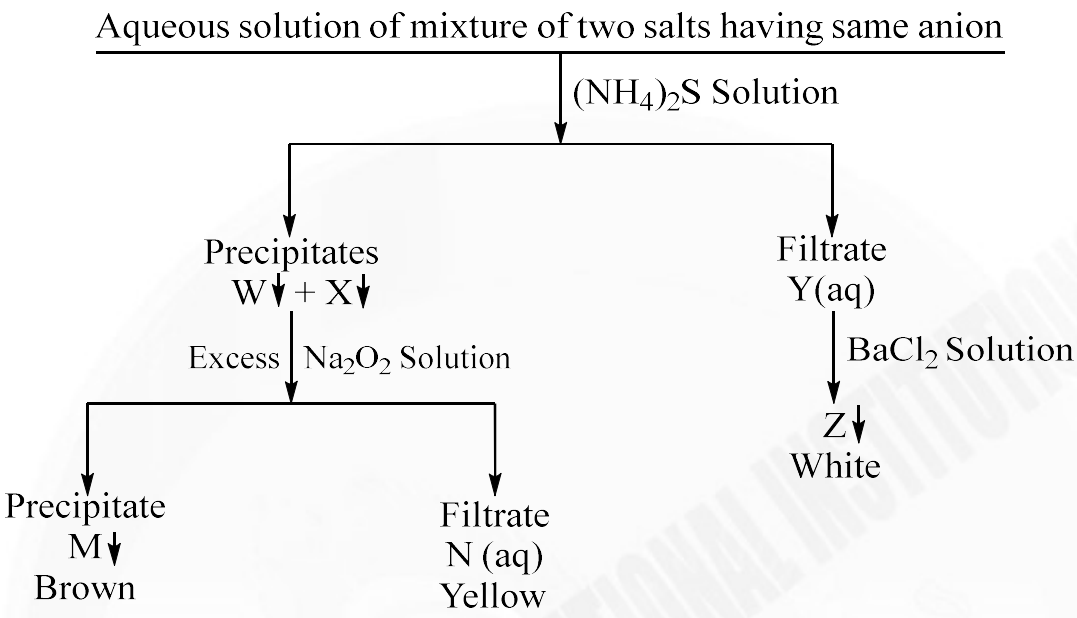
25. In the conversion $Q \rightarrow R$, order of reaction is X. Degree of unsaturation of T is Y. Product of X and Y is

Question Stem for Question Nos. 26 and 27

Question Stem

Consider given series of reactions of aqueous solution of mixture of two water soluble salts having same anion.

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26. If precipitate Z is insoluble in dilute HCl and anionic species N follows given reaction:

$$N(aq) + H^+(aq) + H_2O_2(2\text{-}3\%) \xrightarrow{\text{Amyl Alcohol}} \underset{\text{deep blue}}{P}$$

then find out numerical value of $|a+b|$

where; a = Oxidation number of central atom of anion of the given salts.

b = Total number of lone pair(s) of electrons on atoms having oxidation state -1 in deep blue coloured species P.

27. **i)** $M\downarrow + HCl \xrightarrow{\text{warm}} \underset{\text{yellow}}{Q(aq)}$

ii) $Q(aq) + SCN^-(aq) \longrightarrow \underset{\text{Blood Red Coloured}}{R}$

Find out number of unpaired electrons in complex species R.

Question Stem for Question Nos. 28 and 29

Question Stem

Weak acids and weak bases are weak electrolytes. They do not undergo complete ionisation at high concentration. With the dilution of weak electrolytes degree of ionisation increases but ionisation constants do not change. At infinite dilution degree of ionisation is nearly equal to unity. Take K_a of $CH_3COOH = 10^{-5}$

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28. pH solution of in which CH_3COOH is 50% percentage ionized is

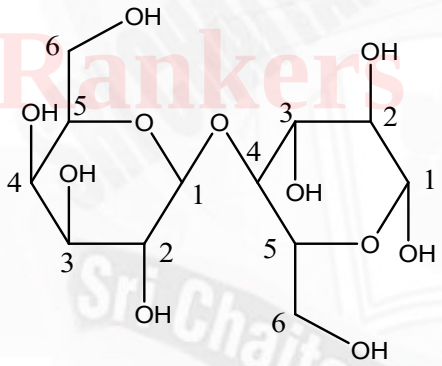
29. Upon diluting 11 L of $0.1M\ CH_3COOH$ solution to $x\times10^{y}\ L, pH$ of solution doubles. Taking x, y to be single digit integers, calculate $\left(\frac{y}{x}\right)$ (neglect H^+ from water for these calculations)

SECTION 3

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks:* +4** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks:* +3** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks:* +2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks:* +1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks:* 0** If unanswered;
- ***Negative Marks:* -2** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

30. Correct statement(s) about the following disaccharide is/are



A) It has $\beta-(1,4)-$ glucosidic linkage

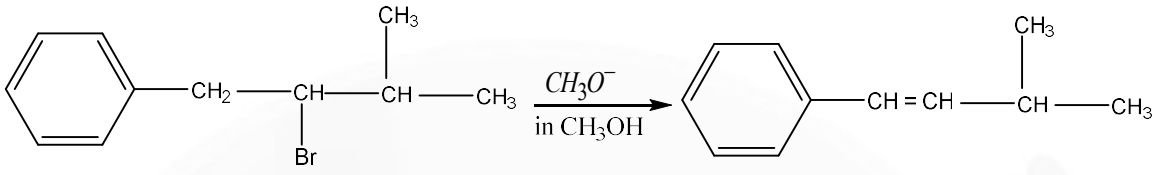
B) When it reacts with excess CH_3OH / HCl gas, molar mass increases by $112\ g/mol$

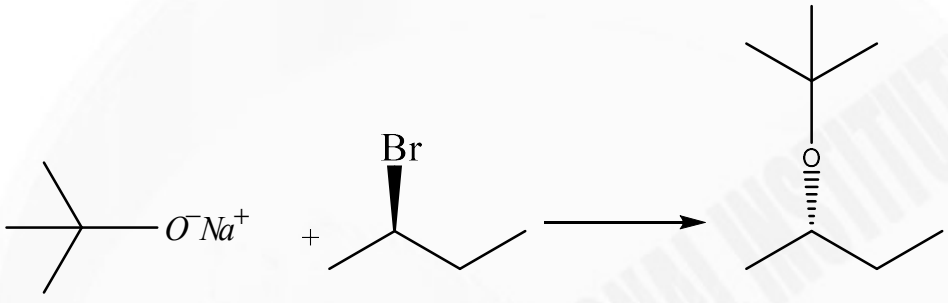
C) It can exhibit mutarotation

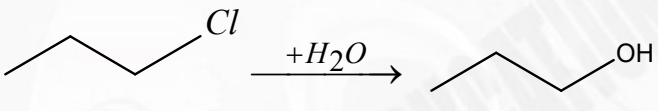
D) one mole of above disaccharide consume maximumly one mole phenyl hydrazine on reaction with excess of phenyl hydrazine

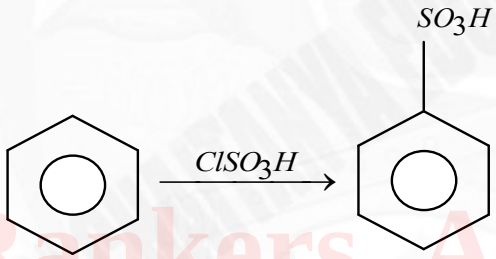
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31. In which of the following reaction(s) correct major product is given ?

A) 

B) 

C) 

D) 

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32. The ionic radii of A^+ and B^- are 0.165 nm and 0.181nm respectively. Their atomic weights are $112u$ and $80u$ respectively. Their binary compounds is known to exist in a type of cubic lattice formed so as to maximise coordination number for the ions. The correct statements for this crystal are ? [Take $\sqrt{2}=1.41$ and $\sqrt{3}=1.73, N_A=6\times10^{23}$]

A) The distance between nearest A^+ ions is 0.4 nm

B) Number of formula units per unit cells is 4.

C) The density of defect-free AB is $5\,gcm^{-3}$.

D) The packing fraction for this structure is more than that for a body centred cubic structure containing identical atoms.

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33. Select the correct statement(s):

A) Surface active agent like soaps and synthetic detergents can form associated colloids.

B) Soaps molecules can act as emulsifying agents.

C) Addition of Sodium Stearate to water decreases surface tension of water till CMC is reached. Beyond CMC however, surface tension does not change much.

D) Micelles formation starts at temperatures just above critical temperature.

34. Consider given reactions:

(i) $SO_3(g) + H_2O \rightarrow X \xrightarrow{SO_3(g)} Y$

(ii) $SO_3(g) + H_2O_2 \rightarrow P \xrightarrow{SO_3} Q$

and choose correct statement(s):

A) All oxyacids produced in given reactions have same basicities.

B) Oxyacid Q can oxidise $Mn^{2+}(aq)$ ions into $HMnO_4$.

C) Oxidation state of sulphur in Y is different from oxidation state of sulphur in P.

D) Maximum number of directly bonded atoms in Q which can lie on straight line are two.

35. Choose correct statement(s) about process of oxidation used in refining of impure metals:

A) It is known by various names like Bessemerisation, Puddling, Softening.

B) It is used when impurities are preferably oxidised by dioxygen than metal to be refined under exposure to air.

C) It is used to refine pig iron into wrought iron.

D) Oxidation of impurities is an exothermic process.

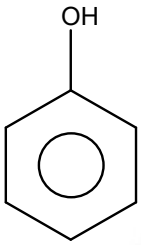
SECTION 4

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

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36.



$\xrightarrow{NaNO_2+Conc.H_2SO_4} P \xrightarrow{diluted\ with\ H_2O} Q \xrightarrow{+NaOH} R$

R is blue coloured anion, Q is red coloured molecule. Sum of degree of unsaturation of Q and R is

37. An equimolar mixture of vapours of $A(l)$ and $B(l)$ is taken in a cylinder-piston arrangement at $200\,torr$ and compressed by gradually increasing external pressure to $600\ torr$. What is the molar ratio of vapours $(A+B)$ to liquids $(A+B)$ at this pressure ? The vapour pressures of pure liquids A and B are 400 torr and 1000 torr respectively. [Assume the piston to be massless]

38. If excess KCN solution is added to an aqueous solution of dipositive metal action having $3d^8$ electronic distribution then find out value of expression $|x+y|$ for resulting cyano complex species.

Where,

x = Number of unpaired electron(s) present in the cyano complex species.

y = Number of electrons present in degenerate 3d orbitals of metal cation having lowest energy level in the cyano complex species.

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MATHEMATICS **Max. Marks: 60**

SECTION 1

- This section contains **Four (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
- *Full Marks* : +3 If ONLY the correct option is chosen;
- *Zero Marks* : 0 If the none of the options is chosen (i.e. the question is unanswered);
- *Negative Marks* : -1 In all other cases.

39. If $f(x)=\begin{cases} x^3+x^2+k & x\in Q \\ x-2 & x\in Q^C \end{cases}$ is continuous at exactly three distinct points then complete set of values of k is

A) $\left(-3,-\frac{49}{27}\right)$ **B)** $\left(-1,\frac{1}{3}\right)$

C) $(-\infty,-3)\cup\left(-\frac{49}{27},\infty\right)$ **D)** $(-\infty,-1)\cup\left(\frac{1}{3},\infty\right)$

40. In $\mathbb{R}^3$, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes $P_1: x+2y-z+1=0$ and $P_2: 2x-y+z-1=0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1. Which of the following points lie(s) on M ?

A) $\left(0,-\frac{5}{6},-\frac{2}{3}\right)$ **B)** $\left(0,\frac{5}{6},\frac{2}{3}\right)$ **C)** $\left(-\frac{5}{6},0,\frac{2}{3}\right)$ **D)** $\left(\frac{1}{3},0,-\frac{2}{3}\right)$

41. In R^2, $line\, L: \vec{r}=\left(-\hat{i}-6\hat{j}\right)+\lambda\left(a\hat{i}+b\hat{j}\right)$, where λ is real parameter and curve $C: y^2=8x$.

If line L touches curve C, then sum of all possible value(s) of $\frac{a}{b}$ is

A) 3 **B)** 6 **C)** -6 **D)** -3

42. L_1 is a tangent drawn to the curve $x^2-4y^2=16$ at $A\left(5,\frac{3}{2}\right)$. L_2 is another tangent parallel to L_1 which meets the curve at B. L_3 and L_4 are normal to the curve at A and B. Lines L_1, L_2, L_3, L_4 forms a rectangle, then the radius of the circle circumscribing the rectangle can be…

A) $\sqrt{\frac{109}{2}}$ **B)** $\frac{\sqrt{109}}{4}$ **C)** $\frac{\sqrt{109}}{3}$ **D)** $\frac{\sqrt{109}}{2}$

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SECTION 2

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +2*** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks: 0*** In all other cases.

Question Stem for Question Nos. 43 and 44

Question Stem

The differentiable function $y = f(x)$ has the property that the chord joining any two points, on the curve $y = f(x), A(x_1, f(x_1))$ and $B(x_2, f(x_2))$ always intersects y-axis at $(0, 2x_1x_2)$. Also given that $f(1) = -1$, then

43. 16 times the maximum value of $f(x)$ is …….

44. $|f(2)| =$ ……..

Question Stem for Question Nos. 45 and 46

Question Stem

Two Arithmetic progression having same number of terms equal to k. The ratio of the last term of the first progression to the first term of the second progression equals the ratio of the last term of the second progression to the first term of the first progression both of which are equal to 4. The ratio of the sum of k terms of the first progression to the sum of k terms of second progression is equal to 2. Let α be the ratio of common difference of the first and the second progressions. Let λ be the ratio of their k^{th} terms.

Then

45. The value of $\alpha + 2\lambda$ is equal to…….

46. The ratio of the first terms of first A.P. to second A.P. is……

Question Stem for Question Nos. 47 and 48

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Question Stem

Consider the function f, g, h as follows $h(x)=\begin{cases}1, & x\in[0,2]\\ x, & x\in(2,4]\end{cases}$

$$g(x)=\int_{x}^{x^2} h(t)\,dt, \quad x\in[0,2]$$

$$f(x)=\int_{0}^{x} g(t)\,dt, \quad x\in[0,2]$$

47. Let the number of discontinuity points of $h(x), g(x), f(x)$ be c_1, c_2, c_3 respectively & let number of non-differentiability points of $h(x), g(x), f(x)$ be d_1, d_2, d_3 respectively then the value of $c_1+c_2+c_3+d_1+d_2+d_3=$ ____

48. Area of the region bounded by $y=g(x)$, x-axis & the lines $x=1, x=2$ is k, where $[k]=$ _____ ([.] GIF)

SECTION 3

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks*** **: +3** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks*** **: +2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks*** **: +1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks*** **: 0** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

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49. Let x be real and if $f(x)=2e^{x}+\frac{3i+1}{-2-2i}e^{ix}+\frac{-3i+1}{-2+2i}e^{-ix}$ then which of the following is/are correct $\left(i=\sqrt{-1}\right)$

A) $f(x)$ is purely real $\forall x\in R$ **B)** $|f(x)|$ is increasing for $x\in\left[0,\frac{\pi}{2}\right]$

C) $\int_0^{\pi/2}|f(x)|\,dx=2e^{\pi/2}-3$ **D)** $\frac{d}{dx}(|f(x)|)$ at $x=\frac{\pi}{4}$ is $2e^{\pi/4}+\frac{3}{\sqrt{2}}$

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50. Consider positive integers $k_1, k_2, k_3, \ldots, k_n$ such that $k_1 + k_2 + k_3 + \ldots + k_n = 5n - 4$ (where $n \in N$) and $\frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \ldots + \frac{1}{k_n} = 1$. Then which of the following is/are correct

A) 'n' can take 3 possible values

B) 'n' can take 4 possible values

C) A possible value of k_n can be 5 for some 'n'

D) Total number of distinct n - tuples $(k_1, k_2, k_3, \ldots, k_n)$ for distinct values of 'n' satisfying the system of equation is 3 .

51. $f(x) = \sin\left(x + \tan^{-1} x\right) + \cos\left(x - \cot^{-1} x\right); x \in R$, then which of the following options is /are correct?

A) $y = f(x)$ is an odd function

B) Range of $y = f(x)$ is $[-2, 2]$

C) The least positive value of ' x ' for which $f(x) = 2$ is α, then $\alpha > \frac{\pi}{4}$.

D) $\left[\int_0^{\pi/4} f(x)dx\right] = 1$ where [.] denotes G.I.F

52. 1,1,2,3,3,4,5,5 are the numbers written on eight separate cards. Three cards are picked at random. Then which of the following options is/are correct?

A) Probability that sum of 3 numbers on the picked cards is odd is $\frac{13}{28}$.

B) Probability that variance of the 3 numbers on the picked cards is the greatest variance possible is $\frac{1}{7}$.

C) Probability that sum of 3 numbers on the picked cards is odd is $\frac{1}{2}$.

D) Probability that variance of the 3 numbers on picked cards is the least variance possible is $\frac{1}{14}$.

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53. Let $f(x)=e^x\sqrt{x}(x>0)$ &$g(x)$ be its inverse. If $I=\int_0^1 f(x)dx$, then which of the following is/are true ?

A) $\int_0^e g(x)dx=e-I$ **B)** $\int_0^1 \frac{f(x)}{x}dx=2(e-I)$

C) $I>1$ **D)** $I<1$

54. If $\frac{\sin\theta}{\cos\theta}+\frac{\sin 2\theta}{\cos^2\theta}+\frac{\sin 3\theta}{\cos^3\theta}+\ldots+\frac{\sin n\theta}{\cos^n\theta}=\cot(p\theta)-\frac{\cos(q\theta)}{\sin^r\theta\cos^s\theta}$, then (Given that $p,q,r,s,(q<0)$ are unique integers) which of the following is/are true?

A) $\text{prs}+1=\text{q}$ **B)** $q^2=s^2+r^2+2ps$

C) $\text{q}+\text{r}+\text{s}=0$ **D)** $s-p,s,q$ are in A.P.

SECTION 4

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

55. If $A_1=[a_1], A_2=\begin{bmatrix}a_2 & a_3\\ a_4 & a_5\end{bmatrix}, A_3=\begin{bmatrix}a_6 & a_7 & a_8\\ a_9 & a_{10} & a_{11}\\ a_{12} & a_{13} & a_{14}\end{bmatrix},\ldots,A_n=[\ldots]$ and so on, where $a_r=[\log_2 r], r\in N$. Where [.] denotes greatest integer function). Then $T_r(A_{10})$ is equal to; (T_r stands for trace of the matrix) ____

56. $a=(Z+2)(\bar{Z}+2i)$ is purely real and $a\in[-1,2]$ where $Z=x+iy, x, y\in R$. then $(|Z|_{\min})^2+(|Z|_{\max})^2$ .....

57. If the quadratic equation $x^2+(2-\tan\theta)x-(1+\tan\theta)=0$ has two integral roots, then sum of all possible values of θ in the interval $(0,2\pi)$, is $k\pi$, the value of k, is

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42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
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BELOW 100 ALL INDIA OPEN CATEGORY RANKS	29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	745	NUMBER OF QUALIFIED RANKS	4,212

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